MarTech Engineer – Assessment

PART 1 — Strategic Assessment

Task 1 — Past Project (Short)

Describe one project where you significantly improved tracking, attribution, or analytics infrastructure.

Analytics Data Governance

While at Spotlight Sports Group, maintaining high data quality was particularly challenging. App feature releases occurred every two weeks and required sign-off on more than 80 tracked events. The process was entirely manual, often taking several days to complete. It was error-prone and frequently delayed releases.

The objectives were to:

1. Reduce sign-off time
2. Increase confidence in data accuracy
3. Identify and resolve issues earlier in the development cycle

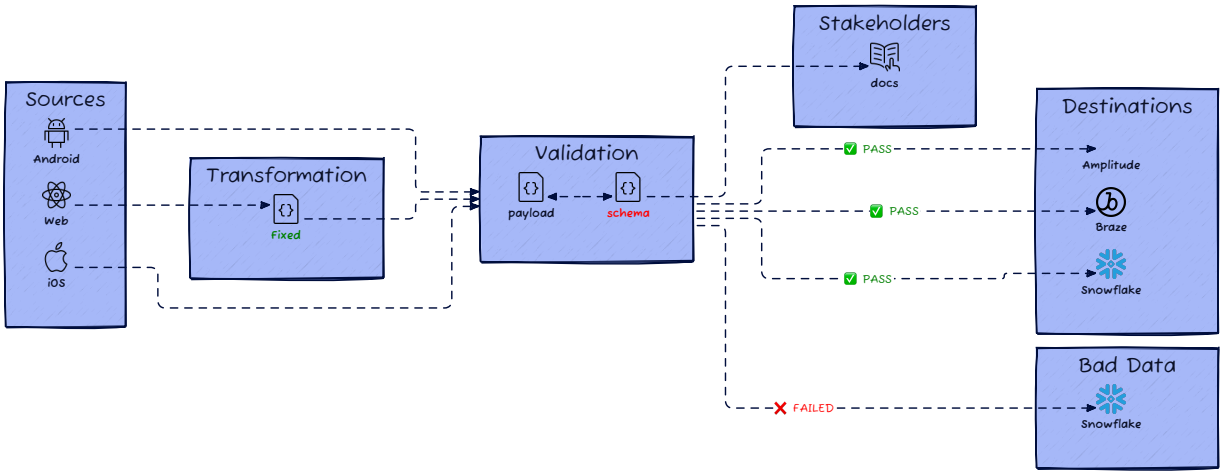
I led the introduction of a real-time and automated validation framework for tracked events by:

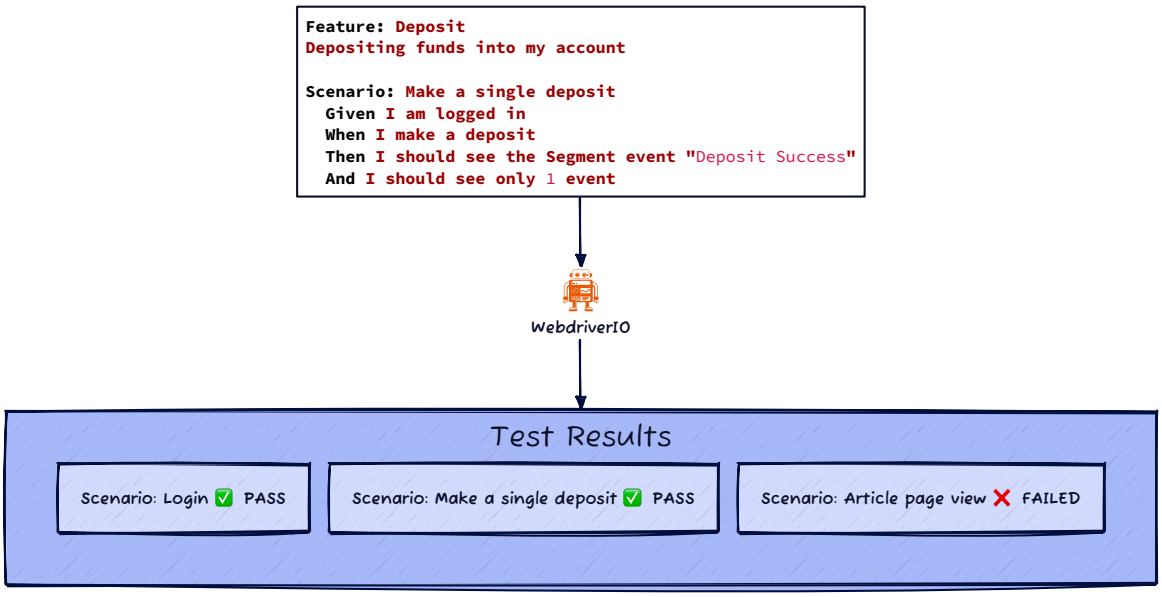
- Isolating tracking implementations to align with the development lifecycle: development, testing, UAT and production.
- Using JSON schemas that evolved with the tracking requirements.
- Leveraging the schemas to produce stakeholder-friendly living documentation.
- Facilitated a mechanism to amend tracking in-flight when fixing issues at the source was not feasible within the sprint timeframe.
- Designing and maintaining end-to-end automated tests to replace manual validation processes.

As a result:

- Analytics sign-off times were reduced by around 50%, with issues identified much earlier in the release cycle.
- Confidence in tracking increased significantly, both internally and among stakeholders.
- A scalable, flexible solution was established, capable of evolving in line with product and brand development.

Figure 1: Tracking plans, transformations, and automation





Task 2 — Attribution & Decision Making

1. *What are the risks of evaluating performance this way?*

   Assessing performance solely using these metrics risks skewing the results towards low-cost, short-term acquisitions, while overlooking the long-term quality and value of the acquired users as in the example.

   A low CPA may indicate cost efficiency, it doesn't account for whether the acquired user will remain engaged or retain value over time.

   Learning phases usually take 7-10 days before they stabilise.

   Similary D7 ROAS would not suitable for awareness campaigns as these are placed at TOFU.

2. *What would you change to measure performance more accurately?*

I would shift focus from short-term vanity metrics to longer-horizon indicators like D30/D90 revenue, actual LTV, and especially predictive LTV (pLTV), which better captures a user's long-term value and relationship with the brand. The latter can be expanded into LTV:CAC ratios.

1. *How would you implement this change at a data level*
   - D30, D90, and LTV are usually provided by most MMP
   - Predictive LTV would require modelling based on first-party data

Task 3 — iOS Attribution (SKAN + AppsFlyer)

1. *How would you integrate this data into internal reporting?*

   1. Leverage Data Locker in AppsFlyer to export data to a storage bucket (e.g.: Google Cloud Storage) or DW (e.g.: BigQuery)

   2. Use Airflow to orchestrate automated workflows that transform the data (e.g.: dbt) into marts

   3. Connect the BI tool (Looker, Power BI, Tableau, etc.) to the marts

2. *How would you avoid double counting?*

   I would leverage AppsFlyer's 'Single Source of Truth' which combines all attribution methods used by the platform.

3. *What are the main pros and cons of:*

   - *SKAN*
   - *Probabilistic attribution*

   SKAN enables campaign tracking on devices where users have opted out of ATT. However, due to its privacy-centric design, SKAN provides aggregated data instead of user-level insights. Moreover delays of two days or longer is expected. Probabilistic attribution may misattribute and over-report. Probability attribution such as the one offered by AppsFlyer still rely on IP addresses and the model are proprietary so the rules of attribution are not disclosed.

4. *How would this affect LTV measurement?*

Since user-level data is unavailable and conversion data is limited and arrives in different windows, LTV values can't be measured directly. Instead, they are predicted based on early user activity.

PART 2 — MarTech Technical Assessment

Container Type	Container Name	Container ID
Server	Brands [Server]	`GTM-PZV4FDG4`
Web	Brands [Client]	`GTM-TSFCWXZZ`

Brand Name	URL	Meta Pixel ID
Brand X	`brand-x-one.vercel.app`	`1533644364407534`

Brand Name	URL	Meta Pixel ID
Brand Y	`brand-y.vercel.app`	`2299157367256743`
Brand Z	`brand-y-zioo.vercel.app`	`1514739266263477`

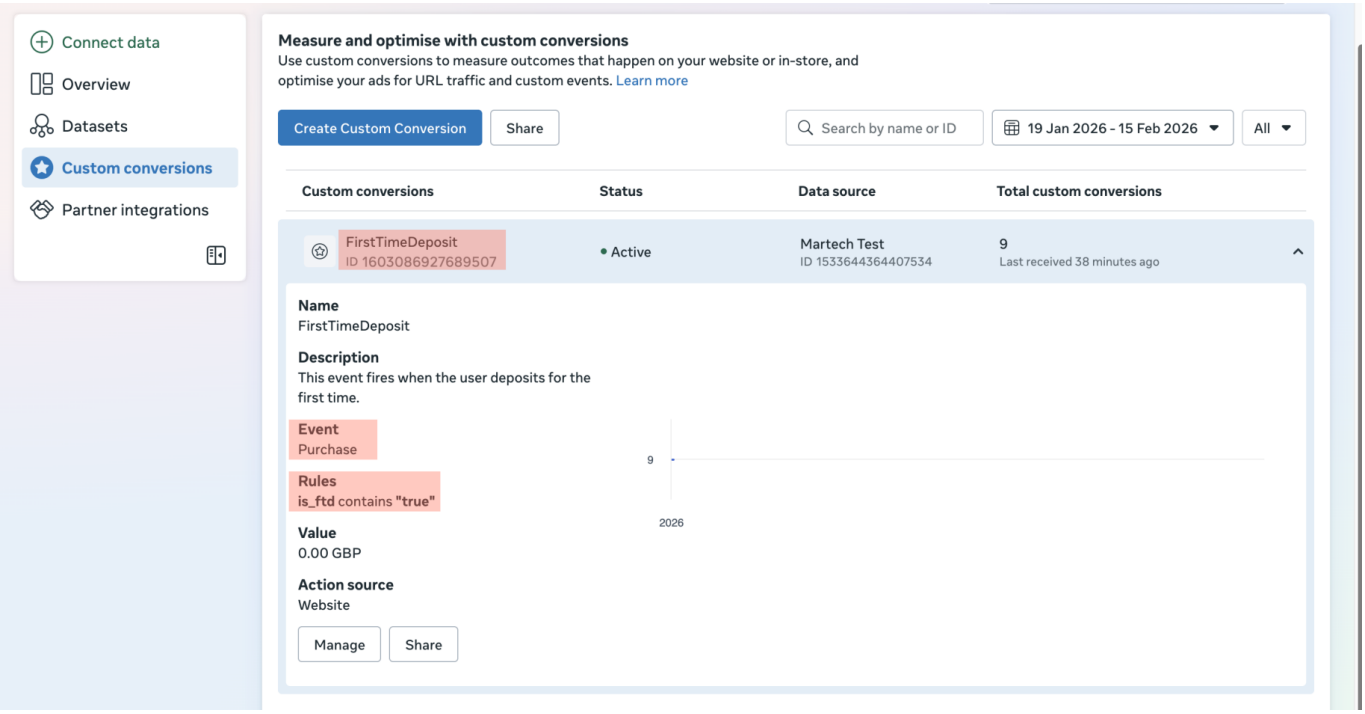
Option A

Track two separate events:

- Standard *Purchase*
- Custom *FirstTimeDeposit*

Option B

Track a single event *Purchase* and derive *FirstTimeDeposit* from *Custom Conversions* in event manager based on the event parameters.




```
// Purchase event sample payload
// Endpoint [POST]: https://server-side-tagging-3phkixd4ta-
uc.a.run.app/collect

{
   "event_name":"purchase",
   "event_id":"266375e4-2ce9-4202-b3d9-3de96a6aa57a",
   "event_time":1771247930,
   "action_source":"website",
   "currency":"GBP",
   "value":"5",
   "is_ftd":"true",
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   "referrer_url":"https://brand-y-zioo.vercel.app/lp/spinners",
   "user_data":{
```

```
      "user_id":"445ef40b-f8e1-4f3f-bb03-f47d93a0875f",
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      "first_name":"First Name",
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      "click_id":"fb.1.1615295627183.1234567890",
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      ],
      "phone":[
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